Session 5 (Live demos): cleaning FLUID-ICU and building Table 1 in R

Foundations of Clinical Research · Week 5, Saturday 19:30 to 22:30 (online) · Neudata · instructor script

	Part A: cleaning	Part B: Table 1 and figures
Slot	5.2 *Meet the course dataset* (about 20:25)	5.4 *Describe the course dataset* (about 21:31)
Duration	8 minutes	6 minutes
Script	`Course/Data/R/01_clean.R`	`Course/Data/R/02_describe.R`
Participants	watch your shared screen and predict in the chat (they do not run R)	same

> **FLUID-ICU is SIMULATED.** Say so every time you open the data: no real patient, hospital or country. All numbers below are from `Course/Data/key_results.md` and `Course/Data/cleaning_log.md`.

Before the session (night before + 10 minutes before)

- [ ] R (4.x) and RStudio installed on the laptop you will screen-share from; packages: tidyverse, janitor, gtsummary, flextable, survival, survminer, mice, broom.
- [ ] Open `Course/Data/FLUID-ICU.Rproj` (sets the working folder). Run `01_clean.R` and `02_describe.R` once end to end. Both must finish without error.
- [ ] Increase the editor and console font (Tools, Global Options, Appearance: 18 to 20 pt) so participants can read it on a small screen. Share only the RStudio window.
- [ ] Keep the fallback figures open in an image viewer: `raw_vs_clean.png` and `table1_snapshot.png` in `Course/Resources/figures/en/data/`, and the five figures that `02_describe.R` saves in `Course/Data/output/figures/` (`fig_shape_age_lactate.png`, `fig_fluid_balance_hist.png`, `fig_sofa_by_group.png`, `fig_infection_source.png`, `fig_los_survivors.png`). The slides of 5.4 show the same figures.
- [ ] Open `Course/Data/cleaning_log.md` and `data_dictionary.md` in a browser tab: you will show them.

Rules of the demo: run one block at a time (Ctrl+Enter). Before each block, ask the class *"what do you expect?"* and read two or three chat answers After it, say *what was fixed and that it goes in the log*. Never type new code live, because the scripts are complete.

Part A: From raw to clean (01_clean.R, 8 minutes)

Step (script block)	What you show	Expected output	Say
0. Read raw as text	`dim(raw)`, `glimpse(raw)`	**623 rows**, 29 columns, everything `<chr>`	"The study enrolled 620 patients. Why 623 rows?"

Step (script block)	What you show	Expected output	Say
1. Column names	`names(dat)`	short lower-case names (`map_mmhg`, `vent_free_days_28`)	"Names without spaces: same in every script."
2. Duplicates	`get_dupes(study_id)`	3 study IDs appear twice, identical rows; after `distinct()` **620**	"Each patient counted once."
3. Sites	`count(site)` before/after	variants such as `icu a`, `ICU-A`: become 4 sites	"Otherwise R sees many different ICUs."
4. Dates	the three formats	`2023-03-14`, `14/03/2023`, `14-Mar-2023`: become one date; 0 failures	"Is 03/04 March or April? Never guess."
5–6. Sex and yes/no	`tabyl()` before/after	many spellings (M, m, male, F, f, " Female"; Y, 1, yes) become `Male/Female`, `Yes/No`	"1 and Yes must mean the same thing."
7–8. Numbers and codes	`summary()` of lactate, MAP, APACHE II	maxima of **999**, minima of **-99**: set to NA	"A lactate of 999 would wreck the mean."
9. Units	lactate median **by ICU**	ICU C about 9 times higher (mg/dL): divided by 9.01	"Found only because we summarised by ICU. Ask the site lab."
10. Impossible values	range checks	age 210, weight 7 kg, a negative fluid output: set to NA + query; patient kept	"We cannot guess 21 or 71. We never delete the patient."
12. Derived variables	the fluid-balance formula	`cum_fluid_balance_d3_ml_kg`, `positive_balance_d3`, `aki_any` computed	"Derived values are calculated, never typed."
13. What is left	`colSums(is.na(dat))`	lactate 93 missing, a handful elsewhere	"These are REAL gaps. We handle them in Session 6."
14. Reproducibility	final `cat()`	`Cleaned data identical to the reference clean file: TRUE`	"Anyone can re-run this and get exactly the same data."

If short of time: run Steps 0, 2, 9, 10 and 14 only (4 minutes) and show the cleaning log for the rest.

Close Part A by opening `cleaning_log.md`: *"Every block you saw is one row here: problem, how many, what we did, why."* Then show the slide *From 623 rows to 527 complete cases* (620 enrolled, then 618 with a day-3 balance, then 527 complete cases).

Part B: Table 1 and figures (02_describe.R, 6 minutes)

Block	What you show	Expected output (key_results.md)	Say
1. Load	`dim(icu), count(positive_balance_d3)`	620 patients, positive balance **373/618 (60.4%)**, 2 without a balance	"618, not 620: two patients have no computable balance."

Block	What you show	Expected output (key_results.md)	Say
2. Shape first	2a: histograms of age and lactate. 2b: mean next to median for each. 2c: the summary that fits	age bell-shaped, mean 62.6 close to median 62. Lactate with a long right tail, mean 2.9 above median 2.6. Then age **62.6 (SD 15.2)**, lactate **2.6 (IQR 1.7 to 3.7)**	"Look first, then choose. Symmetric: mean (SD). Skewed: median (IQR)."
3. Table 1	the gtsummary table by group	negative n = 245, positive n = 373; SOFA **6 (4 to 8) vs 8 (6 to 10)**; vasopressor **24.9% vs 55.0%**; ventilation **39.6% vs 57.9%**; lactate **2.2 vs 2.9** mmol/L; heart failure **22.0% vs 11.5%**; missing counts shown; **no p-values**	"Were the groups alike at the start? No: the positive group was sicker."
4. Histogram	fluid balance by group, with the median and IQR drawn on it	`fb_q` prints the quartiles. On the plot: the black line is the median, **8.8 ml/kg**, the grey band the IQR, **-10.3 to 26.2**, the dashed line the cut at 0	"Continuous exposure, cut at 0."
5. Boxplot	SOFA by group, one dot per patient	`sofa_med` prints the medians. On the plot: **median 6** and **median 8** written next to each box, n under each group	"Show every patient."
6. Bar chart	infection source, sorted	respiratory most common	"Categories: sorted bars, not a 3-D pie."
7. ICU stay histogram	the 439 survivors with a balance (Q5)	median **6.2**, mean **7.5** days, both drawn on the plot	"A few long stays pull the mean up: report the median."
8. Outcomes	described, not compared	28-day mortality **179/620 (28.9%)**, ICU stay in all 620 patients mean 7.3 vs median 6.0 days	"Described today, compared next Saturday."

Close Part B: *"This Table 1 is in every results pack, Q1 to Q5. Now we read it together as a reviewer would."* Then move to the whole-class reading of Table 1 (Exercise 6).

Fallback plan

Problem	Do this
R or a package fails	Show `raw_vs_clean.png` and talk through the problems (Part A). For Part B, show `table1_snapshot.png` and the figures in `Course/Data/output/figures/`, or simply the four result slides of 5.4. Teaching points are identical.
Screen sharing does not show RStudio legibly	Use the PNG figures and `cleaning_log.md` in a browser at 150% zoom.

Problem	Do this
Running late	Part A: Steps 0, 2, 9, 10, 14. Part B: blocks 2, 3 and 5 only.
A participant asks to run it themselves	Point to `Course/Data/README.md` and the free *Clinical Data Analysis in R* course, and offer to help after the session.

Anticipated questions

"Is this real data?" No. It is simulated, designed to behave like ICU data and to show the classic problems.

"Why not fix the errors in Excel?" Hand edits leave no trace and cannot be repeated, and one wrong paste is invisible. A script plus a log is checkable by a reviewer and repeatable by anyone.

"Why set age 210 to missing instead of guessing 21?" It could be 21, 71 or 10, and guessing invents data. Set to missing, query the site, correct only with the source document.

"Why no p-values in Table 1?" Table 1 describes the groups. The SAP says so explicitly. In an RCT any difference is chance. In a cohort it is confounding to be adjusted for, not tested.